

CS 4200 and CS6200

Intro to Quantum Computing

Course Structure

Nella Ludlow, PhD
Wright State University
Asynchronous – Class is ONLINE

CS 4200 / 6200 Overview

- **Course Goals and Learning Outcomes:**
- To understand fundamental terminology and techniques in quantum computing and knowing the current capabilities and limitations of the technology.

This course will cover a broad range of algorithms in depth to include:

- The physics of information processing
- quantum logic
- quantum algorithms
- including Shor's factoring algorithm and Grover's search algorithm
- quantum error correction
- quantum communication and key distribution
- This course will help you establish a foundation of knowledge for understanding what quantum computers can do, how they work, and how you can contribute to discovering new things and solving problems in quantum information science and engineering.

Course Description

Course Description:

- The objective of this class is to provide an ***introduction to quantum computing***. We will cover terminology, brief introduction to quantum mechanics, linear algebra, classical algorithms, understanding of what is a Qubit, 1-qubit operations, and 2-qubit gates, no-cloning theorem, 3-qubit gates, quantum algorithms, the potential of quantum computing to NP and NP-hard problems, code breaking, and optimization.

The overall goal of the course the student should be able to know terminology, understand basic concepts of quantum computing, know challenges to large-scale quantum computing, know the major contributors, and have an ability to read a current journal article in the field.

Students who went to an international conference would understand the talks and know who the leaders in the field were. They could understand, converse, and critique new work and presentations.

Course Goals and Learning Outcomes

Course Goals and Learning Outcomes:

- To understand fundamental techniques in designing algorithms and knowing how to analyze them. This course will cover a broad range of algorithms in depth to include:
 - What is Quantum Computing (QC)
 - Why it is real and Why its time is now
 - Quantum Physics
 - Linear Algebra
 - Quantum Logic
 - Quantum Gates
 - Building Quantum Circuits
 - Introduction to Quantum Algorithms
 - Deutsch-Jozsa Algorithm
 - Simon's Algorithm
 - Grover's Algorithm
 - Shor's Algorithm
 - Code-Breaking
 - Quantum Annealing Machines

Prerequisites

Prerequisite: (Senior or Graduate Student status) and CS3100 Data Structures and MTH 2530 Linear Algebra

- If you're not a Senior or Grad Student email me later. Let's discuss.
- Some students may not have MTH2530--I do each enough Linear Algebra to understand the material. Please note that Quantum Computing and Quantum Algorithms do heavily rely upon Linear Algebra, and the concepts are needed to learn this material.
- I am also a Data Analytics and Mathematics Professor at Washington State University. So, if you have any questions or concerns about learning mathematics, please contact me. I can help you bridge math barriers. Please don't let a fear of math cause you to drop this course. I am confident we can find ways to help you learn this material!

CS6200 – Additional Material

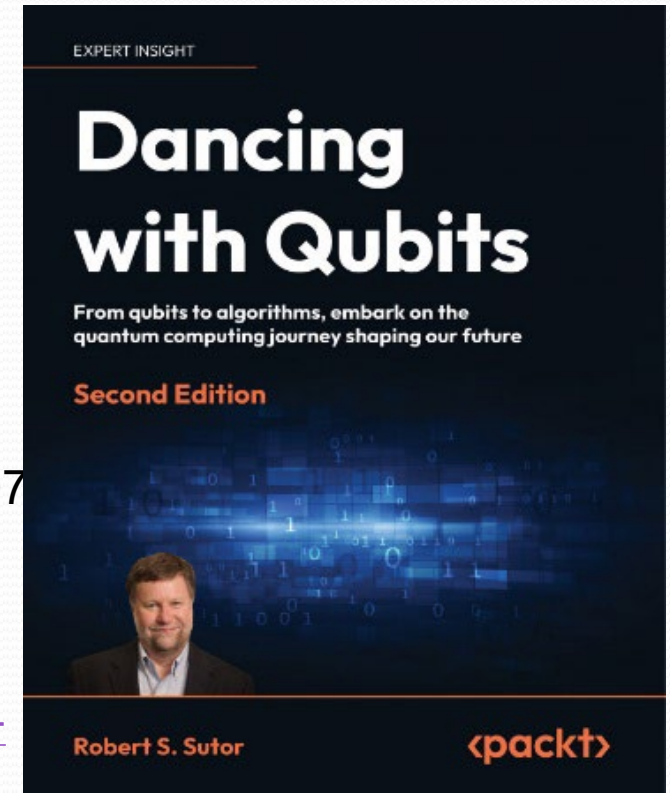
- **Required:** Some additional material for 6000-level Grad Students
- You will be assigned TWO Journal Articles to Review. First one will be assigned around the middle of the semester, and the second one about month later.
- Goal is to understand current journal articles in Quantum Computing. We need to learn some basics first.
- The scores for the Journal Articles will be included in your HW/Assignments grade portion.

Textbook

**Text: Dancing with Qubits,
Robert Sutor, 2nd Ed - 2024.,
Packt> Publishing**

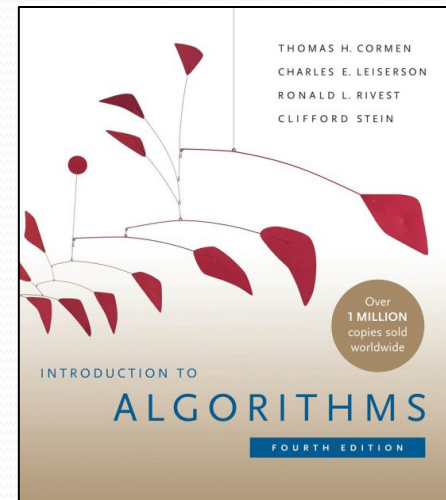
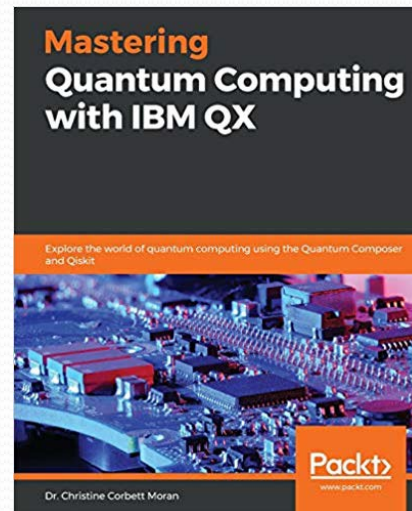
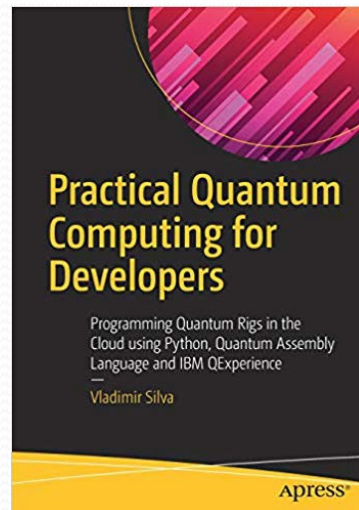
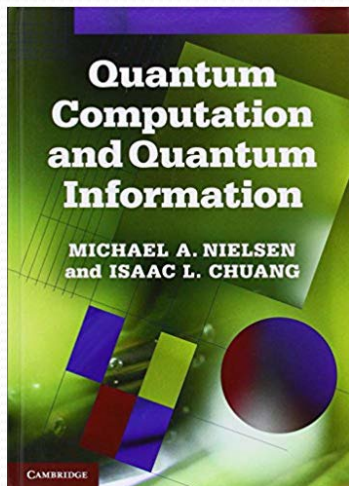
by Robert S. Sutor, 2024, ISBN **978-1837636754**

- Buy anywhere. Amazon Kindle prices is \$44.07
- <https://www.amazon.com/Dancing-Qubits-quantum-computing-change-dp-1837636753/dp/1837636753>
- <https://www.packtpub.com/en-us/product/dancing-with-qubits-9781837636754>



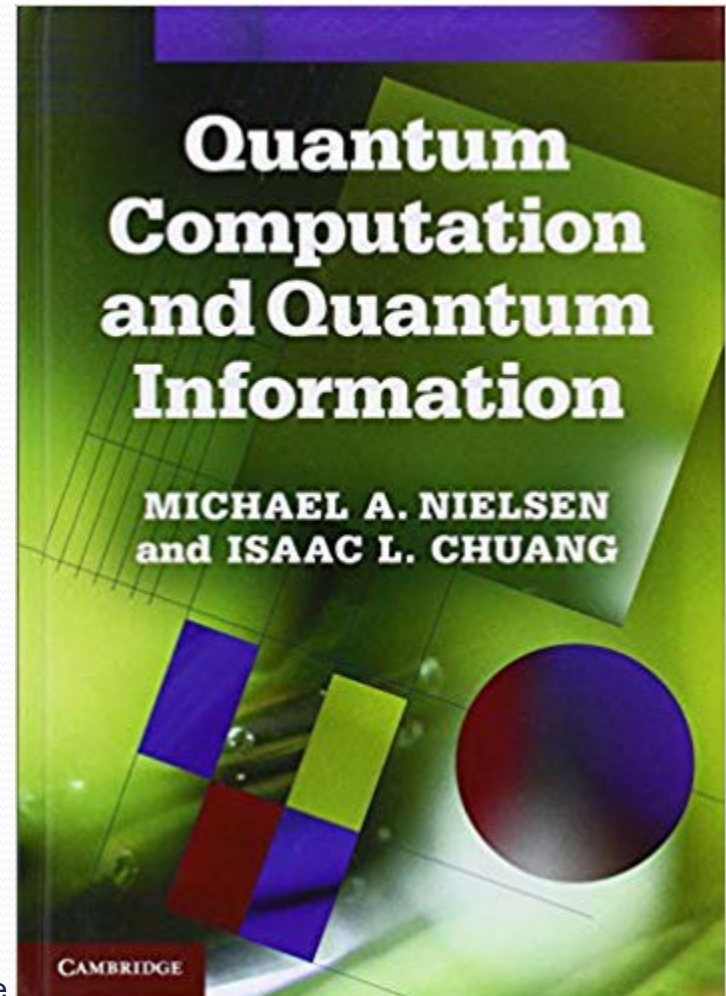
Suggested Books

- These are only suggested – not required



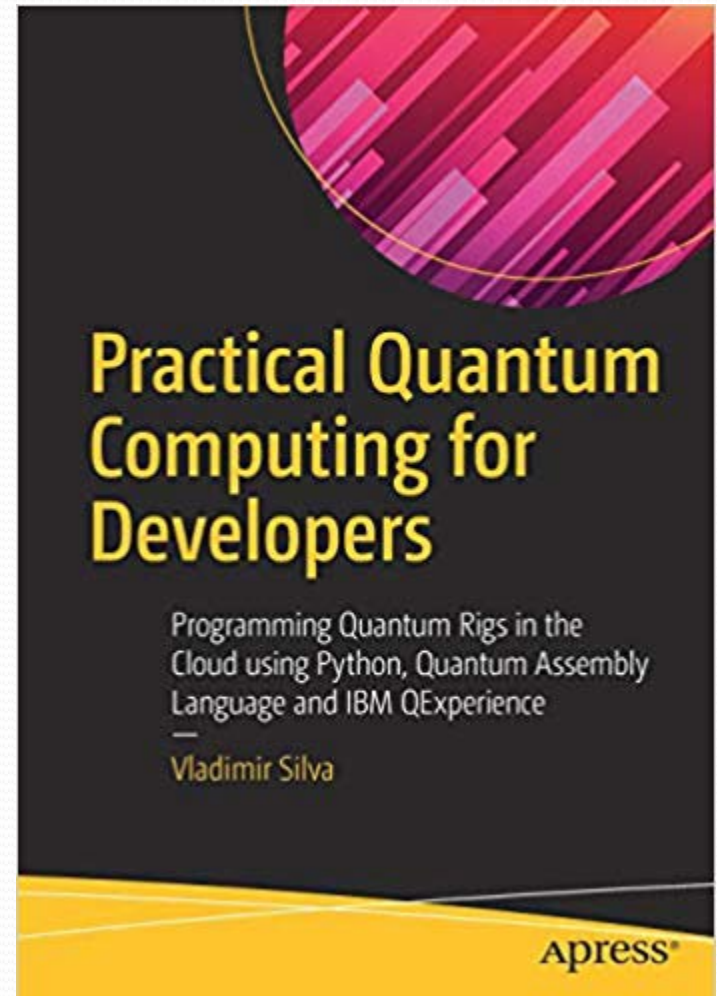
Suggested Textbook

- Text: Quantum Computation and Quantum Information, Nielsen and Chuang, 3rd Ed - 2017., Cambridge Press (ISBN-13: 978-1107002173)
- Its available for \$59 Amazon, \$11 e-format.
- If you buy one book, I recommend you buy this book.
- It is NOT required.



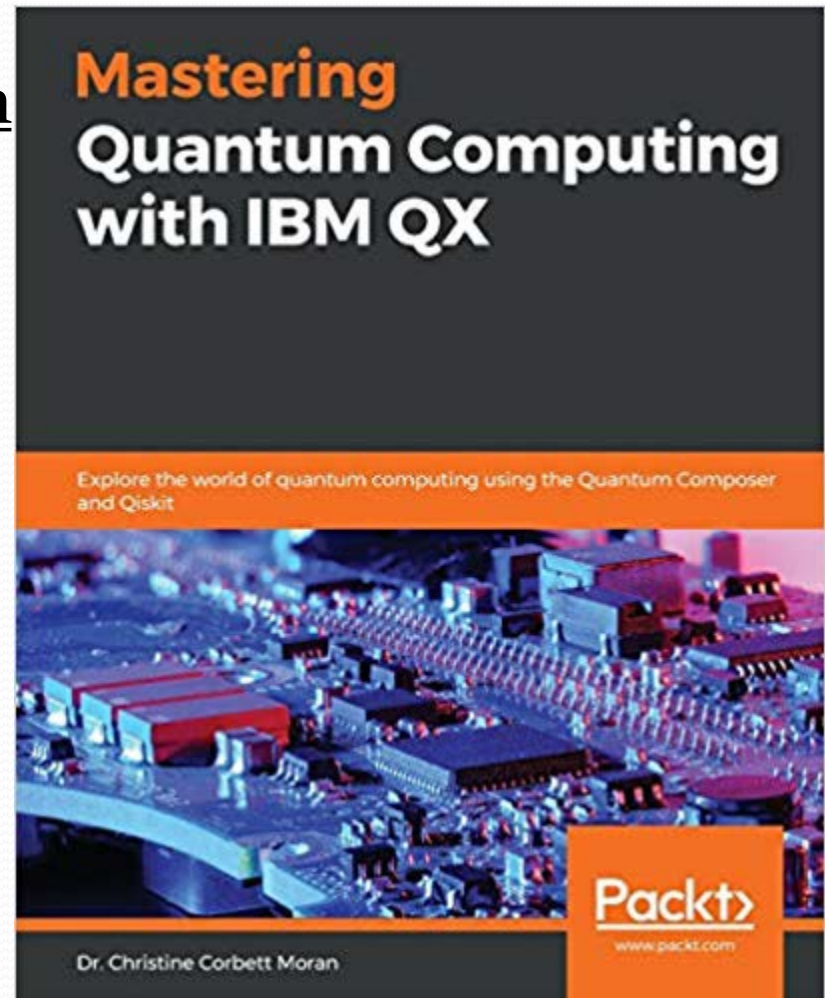
Nice to Have Textbook

- Text: Practical Quantum Computing for Developers, Vladimir Silva, 1stEd - 2018., APress (ISBN-13: 978-1-4842-4217-9-53799)
- Its available for \$24 Amazon, \$19 e-format.
- It is NOT required.



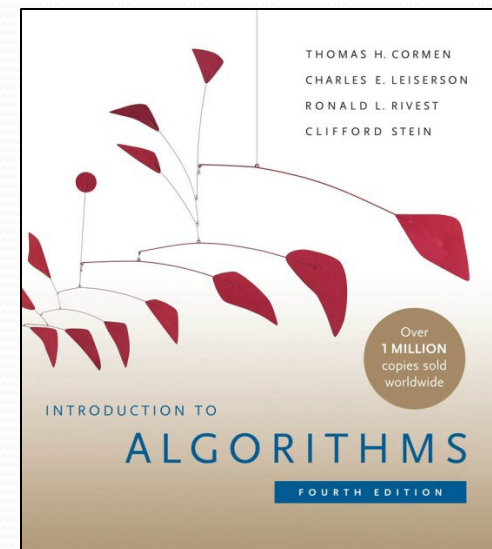
Nice to Have Textbook

- Text: Mastering Quantum Computing with IBM QX, Christine Moran, 1stEd - 2019., Packt (ISBN-13: 978-1-78913-643-2)
- Its available for \$45 Amazon, \$22 e-format.
- It is NOT required.



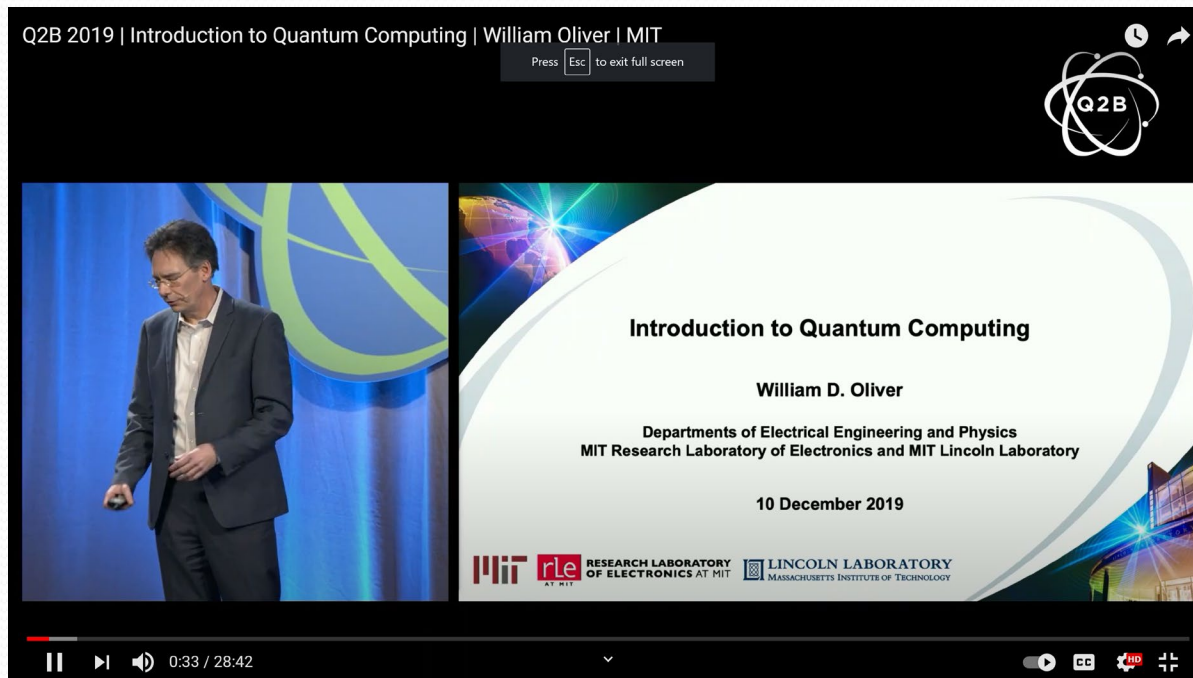
Best Algorithms Textbook

- Text: Introduction to Algorithms, Cormen, Leiserson, Rivest, and Stein, 4th Ed., April 2022, MIT Press (ISBN: 9780262046305) I recommend you buy and keep this book as a computer Scientist.



MIT – William Oliver

- Quantum Computing Certification Courses
- QCWare 2019
- <https://www.youtube.com/watch?v=ZuHHgoe2Boo>

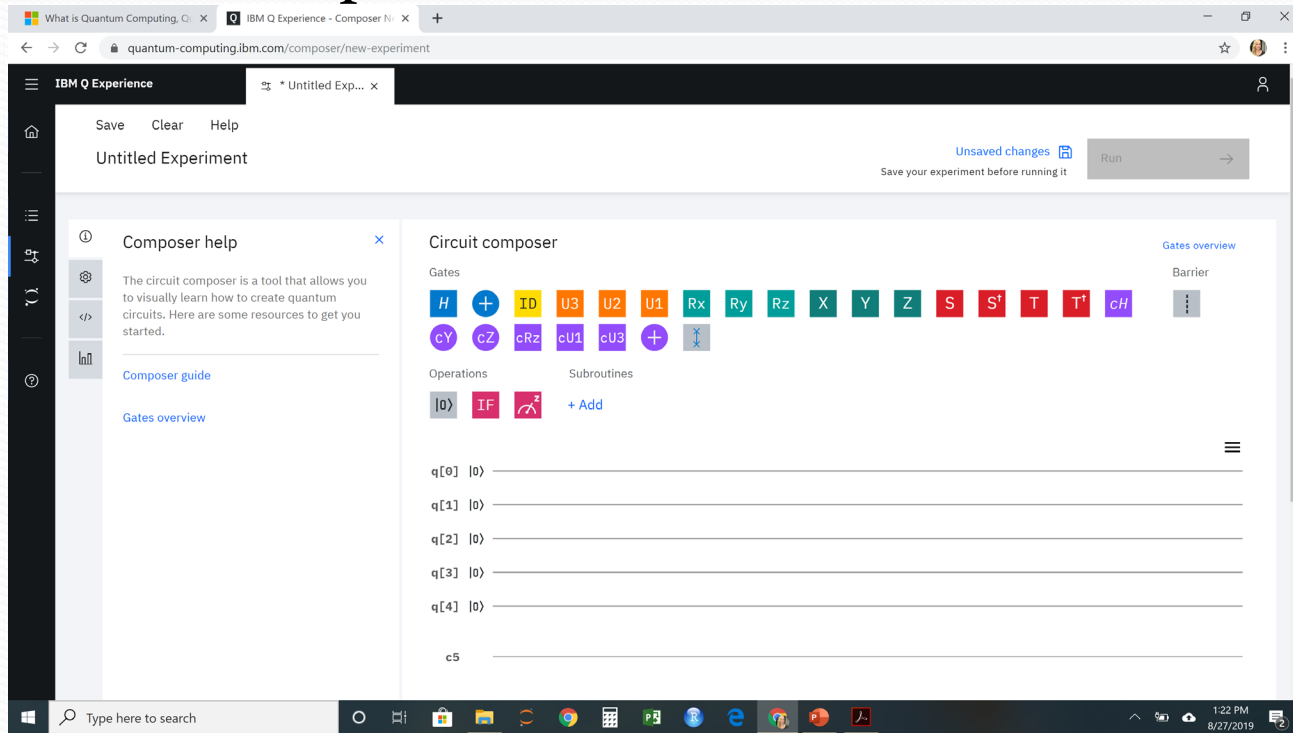


Microsoft Learning Materials

- <https://www.microsoft.com/en-us/quantum/what-is-quantum-computing>
- Krysta Svore Director of Microsoft QuaRC
- Watch Video
- <https://www.microsoft.com/en-us/research/video/krysta-svore-on-quantum-computing-and-machine-learning/>

IBM QX

- <https://www.research.ibm.com/ibm-q/>
- IBM QIS Kit and Circuit Composer
- We will write programs and run them on a REAL Quantum Computer



D-wave

- <https://www.dwavesys.com/resources/videos>
- Quantum Annealing
- <https://youtu.be/MPogiyFIwbw>

Quantum Computing Industry

- Microsoft
- Amazon
- Alibaba
- D-Wave
- IonQ / Rigetti / QC Ware
- Google / IBM
- Cambridge Quantum Computing / Zapata

Quantum in India

- BosonQ.
- Qulabs.ai.
- QpiAI Tech.
- QNu Labs.
- Automatski.
- Quantica Computacao.
- **QRDLab.**
- Taqbit Labs.

Wright State LMS

- *pilot.wright.edu*

Why do I need to check this site?

- Syllabus, Textbook, Files, Lecture Notes
- Frequently Asked (and Answered) Questions (FAQs)
- Upcoming exams and assignments will be posted here
- Misc. static information about the course

How frequently do I need to keep checking the course webpage?

- Twice a week – check any scheduling changes
- You can study materials up two weeks in advance

Grading

- Quizzes 20%
 - Around one every 1-2 weeks
 - All online - submit thru Pilot
- Quantum Program Assignments (two) 20%
- Midterm Exam 1 – week 6 (Thursday) 15%
- Midterm Exam 2 – Week 12 (Thursday) 15%
- Final Exam – Monday of Finals Week 30%

Specific Grading Policy:

- Standard Point Ranges
- Not graded on a curve
- See Syllabus for specifics

Assignment Submission Policy

- Please complete *by midnight of* due date
- *Electronic submission*
- Extensions *may* be permitted under extraordinary circumstances
 - Contact the instructor *ahead of time*
- *I am testing your knowledge of concepts not a specific syntax. This is computer science NOT programming.*
- *Little is little, but nothing is nothing. Turn something in. Let's discuss this, as this is grade killer.*

Academic Policy

- Honor Code. Don't violate this!
 - Collaboration on Homework is fine and encouraged.
 - Assignments/Quizzes/Tests must be done individually, by that I mean:
 - Don't Plagiarize
 - If coding, write your own code!
 - Don't use Chat-GPT or other generative AI
 - Understand what you submit!!!!
 - Don't Cheat on Exams
 - Helping others, getting help, communicating answers -
- I know most of the online sources and often can recognize the material. Do your own work.
- Any deviation from the above rule will be considered cheating and will be subject to the Wright State academic policy

Exam Policy

- 2 Midterms and 1 Final
- Open Book, Open Notes, Open Internet, and is comprehensive
- Make-ups will be rare and only under extraordinary circumstances, if already answers posted
- Seek prior permission from instructor if cannot take on dates scheduled (at least 1 week in advance)

Other Coordination

- Disabilities
 - Other accommodations
 - Classroom Safety
 - Student Attendance
-
- When in doubt, act like this we are at work and we are co-workers. We have different roles, you to learn and for me to teach, but we are on the SAME TEAM!

How To Succeed in this Course

- First, is to Watch Videos
- And, do the reading!
- Turn in your assignments (don't submit a zero)
- The course material builds upon previous material, so don't get too far behind
- Ask Questions
- Anticipate where we are going/ Guess / Predict
- Stay Engaged, Make Comments, Be Active
- Challenge all materials and all professors

I want to help you succeed!

- I am willing to go to extra lengths to help you!
- I hope this course is one of the most important courses you take in your university career
- You put in the energy, and I will help you succeed.
- I welcome feedback and interaction!
- I want to work with you after this class through you graduation! And afterwards too.